

MATH3023 Advanced Mathematics Applications

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Semester 2, 2026

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Preface

THESE NOTES follow the course *Advanced Mathematics Applications*, taught by Lyle Noakes at the University of Western Australia in Semester 2 of 2026. The unit teaches advanced techniques of calculus, including multivariable calculus, Green's, Stoke's and divergence theorem, complex analysis, ordinary and partial differential equations, and Sturm-Liouville theory.

0.1 More detailed overview of the subject.

THESE NOTES are about something but with more detail.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every <i>or</i> for all
$\exists$	There exists
$\in$	In <i>or</i> is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subset of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not <i>or</i> without
$\emptyset$	Empty set
$\sim$	Relates to
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$ n $	The absolute value of n
$\square$	It has been proven

Chapter 1

Review of Multivariable Calculus

1.1 Lecture 1

1.1.1 Double and Triple Integrals

TO INTEGRATE a scalar function $z = f(x, y)$ over a bounded region, we can take either horizontal or vertical slices, and sum them. For example, to calculate $\iint (f(x, y) = xy) dA$ over the region bound by $y = \sqrt{x}$ and $y = x^3$ (fig. 1.1).

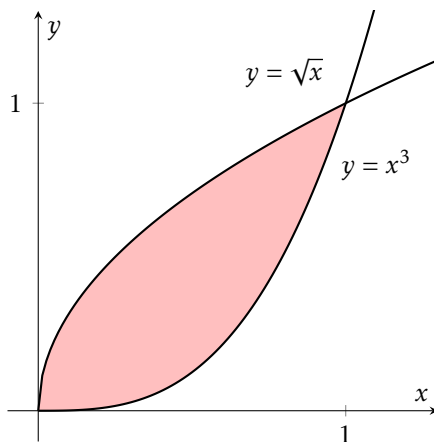


Figure 1.1: The region bound by $y = \sqrt{x}$ and $y = x^3$.

We can either take vertical slices (i.e take regular portions of x and integrate w.r.t y first),

$$\begin{aligned}\int_0^1 \int_{x^3}^{\sqrt{x}} (xy) dy dx &= \int_0^1 \left[\frac{xy^2}{2} \right]_{x^3}^{\sqrt{x}} dx \\ &= \frac{1}{2} \int_0^1 x^2 - x^7 dx = \frac{1}{2} \left[\frac{x^3}{3} - \frac{x^8}{8} \right]_0^1 \\ &= \frac{1}{2} \left(\frac{1}{3} - \frac{1}{8} \right) = \frac{5}{48}\end{aligned}$$

Or we can take horizontal slices (i.e take regular portions of y and integrate w.r.t x first),

$$\begin{aligned}\int_0^1 \int_{x=y^2}^{x=y^{1/3}} (xy) dx dy &= \int_0^1 \left[\frac{x^2 y}{2} \right]_{y^2}^{y^{1/3}} dy \\ &= \frac{1}{2} \int_0^1 y^{5/3} - y^5 dy = \frac{1}{2} \left[\frac{3y^{8/3}}{8} - \frac{y^6}{6} \right]_0^1 \\ &= \frac{1}{2} \left(\frac{3}{8} - \frac{1}{6} \right) = \frac{5}{48}\end{aligned}$$

Both approaches yield the same result, but one may be easier than the other. For example,

$$\int_{y=0}^{y=1} \int_{x=\sqrt{y}}^{x=1} e^{x^3} dx dy$$

is impossible, but reversing the order of integration makes it doable.

$$\begin{aligned}\int_{x=0}^{x=1} \int_{y=0}^{y=x^2} e^{x^3} dy dx &= \int_0^1 \left[e^{x^3} y \right]_0^{x^2} dx \\ \text{let } u &= x^3 : \int_0^1 x^2 e^{x^3} dx = \frac{1}{3} \int_0^1 e^u du \\ &= \frac{1}{3} [e^u]_0^1 = \frac{1}{3}(e-1)\end{aligned}$$

TRIPLE INTEGRALS are solved in much the same way, as a series of single integrals.

$$\begin{aligned}\iiint f(x, y, z) dV : a \leq x \leq b, q(x) \leq y \leq w(x), g(x, y) \leq z \leq h(x, y) \\ = \int_a^b \left(\int_{q(x)}^{w(x)} \left(\int_{g(x, y)}^{h(x, y)} f(x, y, z) dz \right) dy \right) dx\end{aligned}$$

To find the volume of a region R , integrate the function $f(x, y, z) = 1$.

1.1.2 Changing Variables in Double and Triple Integrals

To CHANGE from (x, y) to (u, v) , you first need to define a transformation $\mathbf{g}(u, v) = (x, y)$. A common transformation is from (x, y) to (r, θ) , with r being the distance from the origin, and θ being the angle from an arbitrary axis.

$$\mathbf{g}(r, \theta) = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Then, define the *Jacobian* matrix to be:

$$\mathbf{J} = \frac{\partial \mathbf{g}(u, v)}{\partial (u, v)} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}$$

For the polar coordinates transformation,

$$\mathbf{J} = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}$$

Then,

$$\iint f(x, y) dx dy = \iint f(x(u, v), y(u, v)) |\mathbf{J}| du dv$$

For polar coordinates,

$$|\mathbf{J}| = r \cos^2 \theta + r \sin^2 \theta = r$$

So,

$$\iint f(x, y) dx dy = \iint f(r, \theta) r dr d\theta$$

The process is the same for triple integrals, except that calculating the determinant by hand is slightly more annoying. When

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w)$$

$$\mathbf{J} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{bmatrix}$$

1.1.3 Polar Coordinates

As discussed above, polar coordinates involves expressing points on the Cartesian plane in terms of r and θ , where r is the distance from the origin, and θ is the angle from an axis, usually the x -axis. Switching from $(x, y) \rightarrow (r, \theta)$ is usually beneficial when the problem being solved is symmetric about rotation around some point.

For example, switch from $(x, y) \rightarrow (r, \theta)$ to solve $\iint_R \sin(x^2 + y^2) dx dy$ where R is the region $1 \leq x^2 + y^2 \leq 16$ (fig. 1.2).

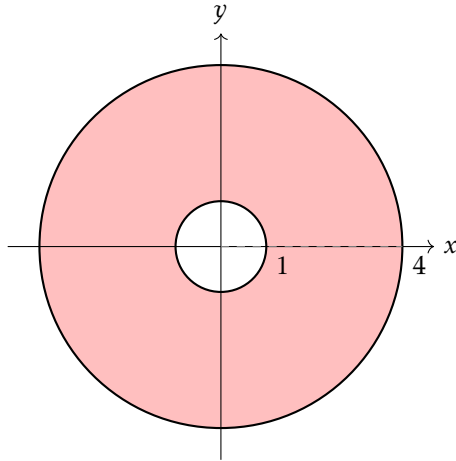


Figure 1.2: Plot of R .

$$\iint_R \sin(x^2 + y^2) dx dy = \int_0^{2\pi} \int_1^4 r \sin(r^2) dr d\theta$$

$$\begin{aligned}
 \text{let } u = r^2, \quad du = 2r \, dr : \quad & \int_0^{2\pi} \int_1^4 r \sin(r^2) \, dr \, d\theta = \frac{1}{2} \int_0^{2\pi} \int_1^{16} \sin(u) \, du \, d\theta \\
 & = \int_0^{2\pi} [-\cos(u)]_1^{16} \, d\theta = [(\cos(1) - \cos(16))\theta]_0^{2\pi} \approx 9.4
 \end{aligned}$$

1.1.4 Cylindrical Polar Coordinates

CYLINDRICAL POLAR COORDINATES involves changing from $(x, y, z) \rightarrow (r, \theta, z)$ (fig. 1.3).

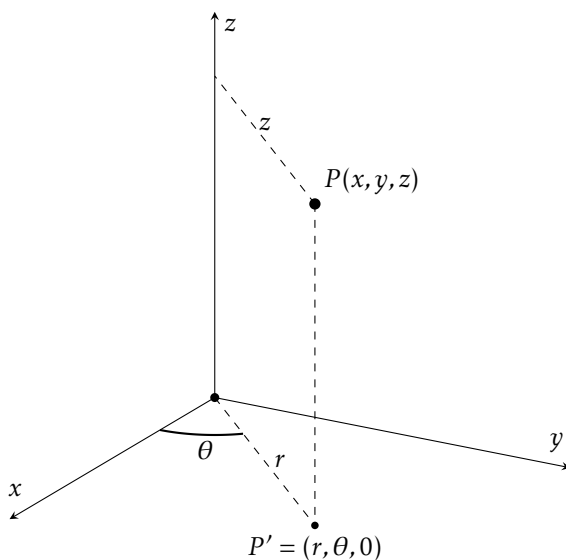


Figure 1.3: Cylindrical Polar Coordinates Projection

The point $P = (x, y, z)$ is projected down to the x, y -plane and then converted to polar coordinates; $P(x, y, 0) \rightarrow P(r, \theta, 0)$. The z -coordinate is then reused. Cylindrical polar coordinates are best used for problems that are symmetric about rotation around the z -axis. Since this is essentially polar coordinates, the determinant of the Jacobian is r .

For example, the mass of a solid cylinder C of radius 1, and height 2, centered

at the origin (fig. 1.4) has density

$$D(x, y, z) = 1 + x^2 + y^2 + z^2$$

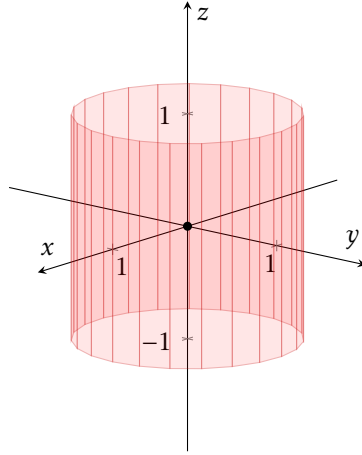


Figure 1.4: Plot of Cylinder C

To find the mass of this cylinder, we need to solve $\iiint_C D(x, y, z) dV$

$$\begin{aligned} \iiint_C D(x, y, z) dV &= \iiint_{C'} D(r, \theta, z) r dr d\theta dz \\ &= \int_{-1}^1 \int_0^{2\pi} \int_0^1 r(1 + r^2 + z^2) dr d\theta dz \\ &= \int_{-1}^1 \int_0^{2\pi} \left[\frac{r^2}{2} + \frac{r^4}{4} + \frac{r^2}{2} z^2 \right]_0^1 d\theta dz = \int_{-1}^1 \int_0^{2\pi} \frac{3}{4} + \frac{1}{2} z^2 d\theta dz \\ &= \int_{-1}^1 \left[\frac{3}{4} \theta + \frac{1}{2} z^2 \theta \right]_0^{2\pi} dz = \int_{-1}^1 \frac{6\pi}{4} + \pi z^2 dz \\ &= \left[\frac{6\pi}{4} z + \frac{\pi}{3} z^3 \right]_{-1}^1 = \frac{11\pi}{3} \end{aligned}$$

1.2 Lecture 2

1.2.1 Spherical Polar Coordinates

SPHERICAL POLAR COORDINATES involves changing from $(x, y, z) \rightarrow (r, \theta, \phi)$, where r is the distance from the point to the origin, θ is the projected angle from the x -axis (the same as cylindrical polars), and ϕ is the angle from the z -axis to the point projected onto the x, z -plane (taking $y = 0$) (fig. 1.5). The determinant of the Jacobian for spherical polar coordinates is $r^2 \sin \phi$.

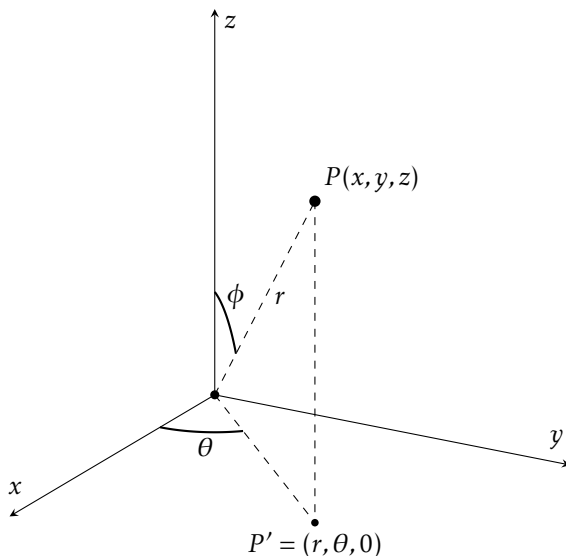


Figure 1.5: Spherical Polar Coordinates

The mass of a sphere of radius 1 centered at the origin with density

$$D(x, y, z) = 1 + x^2 + y^2 + z^2 = 1 + r^2$$

is

$$\begin{aligned} & \int_0^\pi \int_0^{2\pi} \int_0^1 (1 + r^2)(r^2 \sin \phi) dr d\theta d\phi \\ &= \int_0^\pi \int_0^{2\pi} \left[\frac{r^3}{3} + \frac{r^5}{5} \right]_0^1 \sin \phi d\theta d\phi \end{aligned}$$

$$= \frac{8}{15} \int_0^\pi 2\pi \sin \phi \, d\phi = \frac{16\pi}{15} [-\cos \phi]_0^\pi = \frac{32\pi}{15}$$